

Design and Simulation of Unbalanced Mach–Zehnder Interferometer

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Abstract—The main objective of this report is to design and simulate an unbalanced Mach–Zehnder interferometer (MZI) and study how the path-length difference (ΔL) affects its free spectral range (FSR) around 1550 nm. The FSR values obtained from the simulations are then compared with theoretical values calculated using the group index of the waveguide to evaluate the agreement between the simulated and theoretical results.

Keywords: Mach - Zehnder Interferometer (MZI), Free Spectral Range (FSR)

I. INTRODUCTION

The Mach-Zehnder interferometers (MZIs) are important components in silicon photonic circuits and have applications in optical communications, sensing, filtering, and in data centers. The aim of this work is to extract the waveguide group index from the measured free spectral range (FSR) of an unbalanced MZI.

II. THEORY

A. Mach-Zehnder interferometer

Let I_i be the input intensity at the Y-branch of an unbalanced MZI. The Y-branch splits the incoming field equally between the two arms, so $E_1 = E_2 = E_i/\sqrt{2}$. As each field travels along its arm, it picks up a phase, and the fields at the end of the arms become

$$E_{o1} = \frac{E_i}{\sqrt{2}} e^{i\beta_1 L_1}, \quad E_{o2} = \frac{E_i}{\sqrt{2}} e^{i\beta_2 L_2}, \quad (1)$$

where β_1 and β_2 are the propagation constants of the two arms, and L_1 and L_2 are their physical lengths. Around a reference wavelength λ_0 , the propagation constant can be expressed as

$$\beta(\lambda) = \beta_0 + \beta'_0(\lambda - \lambda_0) + \frac{1}{2}\beta''_0(\lambda - \lambda_0)^2 + \dots \quad (2)$$

The output intensity at the combiner Y-branch is written as,

$$I_o = \frac{I_i}{2} [1 + \cos [\beta_1 L_1 - \beta_2 L_2]] \quad (3)$$

For an ideal Y-branch combiner $E_o = \frac{1}{\sqrt{2}}[E_{o1} + E_{o2}]$, neglecting the propagation losses, the output intensity will be;

$$I_o = \frac{I_i}{2} [1 + \cos [\beta_1 L_1 - \beta_2 L_2]] \quad (4)$$

The effective refractive index of both arms are the same, thus $\beta_1 = \beta_2 = \frac{2\pi}{\lambda} n_{eff}$. Thus the output intensity of an

imbalanced MZI is dependent on the phase difference between the two arms.

$$\Delta\Phi = \frac{2\pi}{\lambda} n_{eff} [L_1 - L_2] \quad (5)$$

Therefore, the transfer function $T(\lambda)$ can be written [1]

$$T(\lambda) = \frac{I_o}{I_i} = 1 + \cos\left[\frac{2\pi}{\lambda} n_{eff} \Delta L\right] \quad (6)$$

B. Extraction of group index from Free Spectral Range (FSR)

The free spectral range (FSR) can be defined as the separation between two maxima. Now if we consider two maxima at λ_{m-1} and λ_m , therefore the phase relations of $m-1$ and m are given below,

$$\Delta\Phi_{m-1} = \frac{2\pi}{\lambda_{m-1}} n_{eff} \Delta L = 2(m-1)\pi \quad (7)$$

$$\Delta\Phi_m = \frac{2\pi}{\lambda_m} n_{eff} \Delta L = 2m\pi \quad (8)$$

The difference between $\Delta\Phi_{m-1}$ and $\Delta\Phi_m = 2\pi$.

The phase change can be written as in Taylor series which is as follows,

$$\Delta\Phi(\lambda) = \Delta\Phi_0 + \Delta\Phi'_0(\lambda - \lambda_0) + \frac{1}{2}\Delta\Phi''_0(\lambda - \lambda_0)^2 + \dots \quad (9)$$

The group refractive index is written as,

$$n_g = n_{eff} - \lambda \frac{dn_{eff}}{d\lambda} \quad (10)$$

Substituting equation 9 in equation 8, and comparing it with equation difference between $\Delta\Phi_{m-1}$ and $\Delta\Phi_m$

$$\frac{2\pi\Delta L n_g}{\lambda_0^2} \Delta\lambda = 2\pi \quad (11)$$

From equation 10, we can calculate n_g as [1],

$$n_g = \frac{\lambda_0^2}{\Delta L \Delta\lambda} \quad (12)$$

where $\Delta\lambda$ is the Free Spectral Range (FSR).

TABLE I
WAVEGUIDE PARAMETERS USED IN THE SIMULATION.

Parameter	Value
Waveguide material	Silicon (Si)
Cladding material	Silicon dioxide (SiO ₂)
Waveguide width	0.500 μm
Waveguide height	0.220 μm
Polarization	Quasi-TE

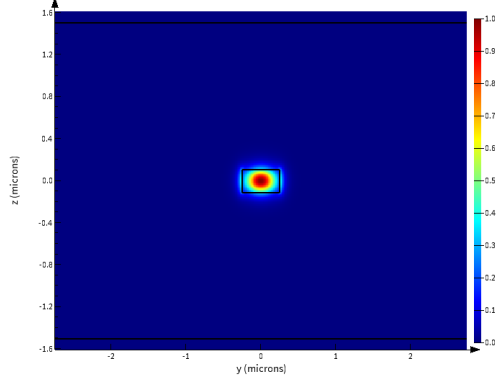


Fig. 1. Lumerical FDE simulated quasi-TE mode profile of the silicon waveguide at 1550 nm.

III. MODELING AND SIMULATIONS

A. Waveguide modelling

The mode profile is simulated at $1.55\mu\text{m}$ and the profile is shown in the figure 1, The effective index at $1.55\mu\text{m}$ is $n_{\text{eff}} = 2.444353$ and group index n_g is 4.207933.

In Lumerical MODE, frequency analysis have been used from which the variation of effective refractive index with wavelength and variation of group index with wavelength is determined across $1.5\mu\text{m} - 1.6\mu\text{m}$ figure 2 and figure 3 respectively.

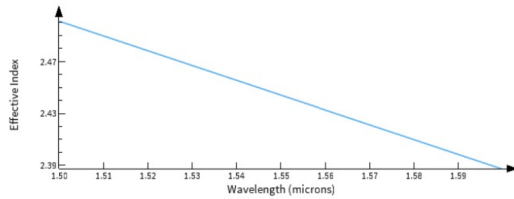


Fig. 2. Simulated effective index for quasi-TE mode from $1.5\mu\text{m} - 1.6\mu\text{m}$

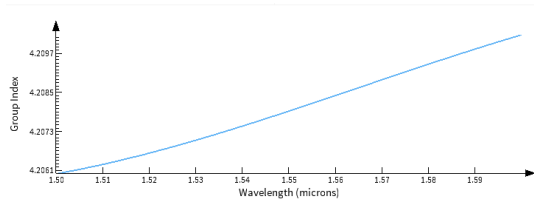


Fig. 3. Simulated group index for quasi-TE mode from $1.5\mu\text{m} - 1.6\mu\text{m}$

For the circuit-level analysis in Lumerical INTERCONNECT, the wavelength variation of the waveguide effective index was approximated by a quadratic model. The expansion was referenced to the design wavelength of $\lambda_0 = 1.55\mu\text{m}$ and written as

$$n_{\text{eff}}(\lambda) = n_0 + n_1(\lambda - \lambda_0) + n_2(\lambda - \lambda_0)^2$$

with wavelength measured in micrometers. The coefficients were determined by fitting the effective-index values extracted from the Lumerical FDE solver across the $1.50\text{--}1.60\mu\text{m}$ wavelength interval. This fitting provides the following compact representation for use in INTERCONNECT:

$$n_{\text{eff}}(\lambda) = 2.444353 - 1.137818(\lambda - 1.55) - 0.0144963(\lambda - 1.55)^2. \quad (13)$$

At the reference wavelength, the FDE calculation gives an effective index of 2.444353. The corresponding group index evaluated at $\lambda = 1.55\mu\text{m}$ is 4.207933.

B. FSR-based group-index extraction

The FSR was verified using the unbalanced MZI model in Lumerical INTERCONNECT. The shorter arm was fixed at $L_2 = 100\mu\text{m}$, while the length of the longer arm, L_1 , was varied from 125 to 200 μm in steps of 10 μm . This produced path-length differences of 25, 35, 45, 55, 65, 75, 85 and 100 μm , respectively.

The FSR was obtained from the simulated transmission spectrum for each arm-length combination is given below

TABLE II
SIMULATED FSR AND EXTRACTED GROUP INDEX FOR DIFFERENT MZI PATH-LENGTH DIFFERENCES.

L_1 (μm)	L_2 (μm)	ΔL (μm)	FSR (nm)	Extracted n_g
125	100	25	22.838	4.294
135	100	35	16.313	4.207
145	100	45	12.688	4.207
155	100	55	10.381	4.207
165	100	65	8.784	4.209
175	100	75	7.613	4.207
185	100	85	6.717	4.207
200	100	100	5.709	4.207

In order to validate the derived relationship for the calculation of FSR in equation 11, the expected FSR was initially calculated analytically based on the group index obtained from the FDE simulation, $n_g = 4.207933$, at the nominal wavelength of $1.55\mu\text{m}$. The path-length difference ΔL was increased from 25 to 100 μm , depending on the combinations of arm-lengths for MZI considered in the INTERCONNECT simulations as shown in figure 4. The results for the calculated FSR were compared to the actual FSR obtained from the simulations. It can be seen that both calculations give quite similar results for all five cases of ΔL . As expected, the increase in the ΔL leads to the decrease in the FSR. Then, the value of group index was estimated from each of the INTERCONNECT-obtained FSR based on the above-derived relationship.

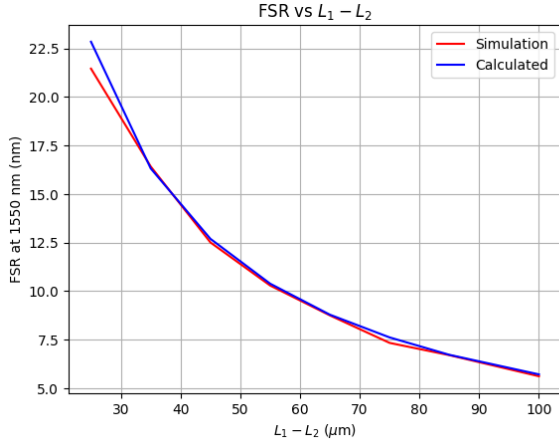


Fig. 4. Comparison of the calculated and simulated free spectral range (FSR) as a function of the MZI path-length difference, ΔL .

TABLE III
COMPARISON OF CALCULATED AND INTERCONNECT-SIMULATED FSR VALUES.

ΔL (μm)	FSR_{calc} (nm)	FSR_{sim} (nm)
25	22.838	21.4523
35	16.313	16.4135
45	12.688	12.5024
55	10.381	10.285
65	8.784	8.747
75	7.613	7.331
85	6.717	6.699
100	5.709	5.6129

C. MZI simulation

To investigate the dependence of the free spectral range (FSR) on the path-length difference, an unbalanced Mach-Zehnder interferometer (MZI) was simulated in Lumerical INTERCONNECT.

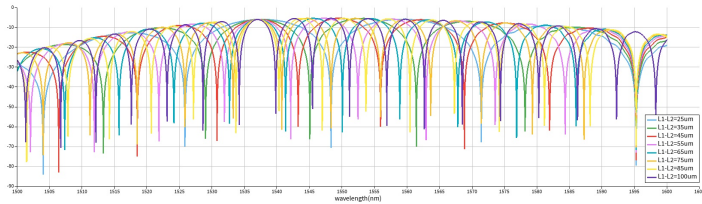


Fig. 5. Optical spectra of unbalanced MZIs with different ΔL ($500 \text{ nm} \times 220 \text{ nm}$ strip waveguide, TE).

As expected, the FSR decreases with increasing ΔL as shown in 3. The extracted group indices range from 4.37 to 4.21, with an average of 4.29.

D. Layout

The mask layout (Fig. 6) was drawn in KLayout using SiEPIC-Tools with the SiEPIC EBeam PDK. It contains eight unbalanced Mach-Zehnder interferometers (MZIs) with path-length differences ΔL ranging from 25 to 100 μm in 10 μm steps, along with a calibrator of 183 μm (-5.12 dB at 1548.19 nm). All devices are built using

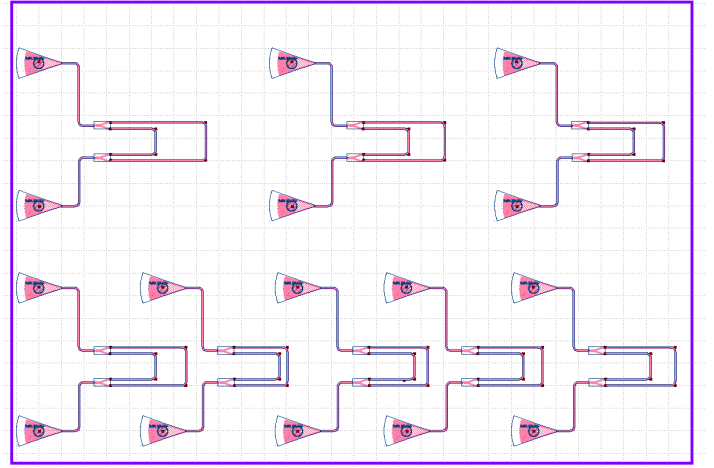


Fig. 6. Optical spectra of unbalanced MZIs with different ΔL ($500 \text{ nm} \times 220 \text{ nm}$ strip waveguide, TE).

the same $500 \text{ nm} \times 220 \text{ nm}$ strip waveguide, Y-branches, and grating couplers (GCs). Each MZI is connected to an input and an output GC, separated by 127 μm , allowing probing with a standard fiber array. The shorter arm was fixed at $L_2 = 100 \mu\text{m}$ for every design, while only L_1 was varied. Each device carries a text label on its input GC for identification during measurements, for example, `opt_in_TE_1550_device_tonijose_MZI<id>`, where the device ID ranges from 1 to 8.

E. Layout Verification

The layout was verified using SiEPIC-Tools (v0.5.31) in KLayout in two steps. First, the KLayout design rule check (DRC) was performed on the final GDS file using the SiEPIC-EBeam-PDK rule deck. The DRC covers boundary, silicon (Si), silicon nitride (SiN), and metal width and spacing rules, and reported no violations in any category. Second, the SiEPIC functional layout check was performed to verify waveguide connectivity and consistency between the layout components and the extracted netlist. The check reported no layout errors. Together, these results confirm that the layout satisfies the specified design rules and connectivity checks before circuit netlist extraction and simulation.

IV. REFERENCES

REFERENCES

- [1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems*. Cambridge, U.K.: Cambridge Univ. Press, 2015.